$$\Rightarrow \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{1}{j\omega}\, d\omega \Rightarrow \frac{2\pi}{2\pi j\omega}$$

$$u(t) = u(t) - u(t)$$

$$H(s) = \frac{V_2(s)}{V_1(s)} = \frac{\frac{R_2}{R_2+L}}{R_1+\frac{R_2}{R_2+L}} = \frac{1}{1+\frac{R_1}{R_2}} - \frac{1}{1+\frac{R_2}{R_1}}$$

2) $i_L = i_C \quad V_L = V_C$

$$V = \frac{1}{C}\int i\,dt$$

$$\int \frac{1}{C}\left(\frac{1}{L}\int V\,dt\right) = \int i\,dt$$

$$\frac{d^2 i}{dt^2} + \frac{1}{LC} i = 0$$

$$i(t) = i(0)\, e^{j\frac{t}{\sqrt{LC}}} + e^{-j\frac{t}{\sqrt{LC}}} \quad \Big| \; i(0) = 1$$

$$V_C(t) = -i(t)$$

$$V_C(s) = V_1(s)\cdot H(s)$$

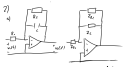
$$= \frac{1}{s} \cdot \frac{\frac{1}{s+\frac{1}{RC}}}{1+\frac{1}{s+\frac{1}{RC}}}$$

$$= \frac{1}{s+\frac{1}{RC}}$$

$$V_C(t) = e^{-\frac{t}{RC}} - e^{-\frac{t}{RC}}$$

3)

a)



$$Z_B \| Z_C = \frac{1}{\frac{1}{R_1}+sC} = \frac{R_1}{1+sR_1C} = \frac{R_1 \frac{1}{C}}{s R_1 + \frac{1}{C}}$$

$$V_1(s) = \frac{Z_{B\|C}}{Z_{B\|C}+R_2}\cdot V_0(s)$$

$$H(s) = \frac{\frac{R_1}{1+sR_1C}}{R_2} = \frac{R_1}{R_2(1+sR_1C)}$$

$$\Rightarrow -\frac{1}{(R_2+C)R_1} = \frac{1}{(R_1C)}\cdot\frac{1}{1+R_2C}$$

4) $v_0(t) = u(t)$

$$V_0(s) = \frac{1}{s}$$

$$V(s) = V_0(s)\cdot H(s)$$

$$= \frac{1}{s}\cdot s\cdot H\left(\frac{1}{s}\right)\cdot\frac{1}{1+\frac{1}{s}}$$

$$= \left(\frac{1}{\frac{1}{s}L\cdot C\cdot(1+\frac{1}{s})}\right)$$

$$\Rightarrow -\left(\frac{1}{LC}\cdot\frac{1}{1+\frac{1}{s}}\right)$$

$$\frac{1}{1+\frac{1}{s}} = A(1+\frac{1}{s}) + B\,s + C$$

$$1 = \frac{1}{s} \Rightarrow A = RLC \quad \Big| \; s=0$$

$$1 = -\frac{RLC}{RLC} \quad \Big| \; s = -\frac{1}{RC}$$

$$B = -\frac{RL}{R}$$

$$\Rightarrow -\left(\frac{RLC}{1+s} - \frac{RL}{1+\frac{1}{s}}\right)$$

$$\Rightarrow \frac{RL}{s}\left(-\frac{s}{1+s} + \frac{1}{1+\frac{1}{s}}\right)$$

$$\Rightarrow -\frac{RL}{s}\left(1 - e^{-\frac{t}{RC}}\right)u(t) \Rightarrow \text{wrt}$$

c) Ideal opamp pi i power = infinite
opp BFF closing loop, so the inverter.

5)

a)

$$Z_B \| Z_C = \frac{R_1}{1+sR_1C}$$

$$V_1(s) = H(s)\cdot V_0(s)$$

$$H(s) = \frac{V_1(s)}{V_0(s)} = \frac{Z_B \| Z_C}{R_2 + Z_B \| Z_C}$$

$$Z_B \| Z_C = \frac{\frac{1}{s}\cdot\frac{1}{sC}}{\frac{1}{s}+\frac{1}{sC}} = \frac{\frac{1}{s}}{s+\frac{1}{C}} = \frac{1}{s^2+\frac{1}{C}}$$

$$\frac{\frac{1}{s^2+\frac{1}{C}}}{R+\frac{1}{s^2+\frac{1}{C}}} = \frac{1}{R(s^2+\frac{1}{C})}$$

$$= \frac{\frac{1}{RC}}{s^2+\frac{1}{RC}s+\frac{1}{LC}}$$

b) $\frac{1}{RC}$

$$\Rightarrow s^2 + 1000s + 10^6 = \frac{1}{s}$$

$$\Rightarrow \frac{-1000s^2}{s^2+1000s+10^6}$$

$$s = \frac{-1000 \pm \sqrt{1000^2 - 4\cdot 10^6}}{2}$$

$$s_1 = -500 + 500\sqrt{3}i$$

(complex) (Conjugate) $P_1 = P_2^* \Rightarrow A_1 = A_2^*$

$$\frac{1000s^2}{(s-500(1+\sqrt{3}i))(s-500(1-\sqrt{3}i))} = \frac{A_1}{(s-500(1+\sqrt{3}i))} + \frac{A_2}{(s-500(1-\sqrt{3}i))}$$

$$\frac{1}{1000} = A_1(1-500(1+\sqrt{3}i)) + A_2(1-500(1-\sqrt{3}i))$$

$$\frac{1}{1000} = A_1(-1000(1+\sqrt{3}i))$$

$$A_1 = -\frac{1}{10^6(1+\sqrt{3}i)} \Rightarrow A_2 = -\frac{1}{10^6(1-\sqrt{3}i)}$$

$$V_0(t) = -e^{500t}\left(A_1 e^{\sqrt{3}it} + A_2 e^{-\sqrt{3}it}\right)$$